IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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<29-04-2026>

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
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Executive Summary

The following methodologies were used to analyze data:

- Data Collection using web scraping and SpaceX API;
- Exploratory Data Analysis (EDA), including data wrangling, data visualization and interactive visual analytics;
- Machine Learning Prediction.

Summary of all results

- It was possible to collected valuable data from public sources;
- EDA allowed to identify which features are the best to predict success of launchings;
- Machine Learning Prediction showed the best model to predict which characteristics are important to drive this opportunity by the best way, using all collected data.

Introduction

Background

SpaceX launches its Falcon 9 rocket for only $62 million, which is much cheaper than other companies that charge over $165 million. The main reason for this big price difference is that SpaceX is able to reuse the rocket's first stage after each launch. So, if we can predict whether the first stage will successfully land and be reused, we can estimate how much a launch will actually cost. This is especially useful for other companies that want to compete with SpaceX when bidding for rocket launch contracts.

What We Want to Find Out

- What are the key factors that decide whether a rocket will land successfully?
- How do different variables and conditions interact with each other to affect the landing outcome?
- What specific conditions need to be met to give the rocket the best chance of landing successfully?

Goal

The main goal of this project is to build a machine learning model that can predict whether SpaceX's Falcon 9 first stage will land successfully or not, based on the available data.

Section 1

Methodology

Methodology

Executive Summary

- This project analyzed SpaceX Falcon 9 launch data to predict whether the first stage would land successfully. Here's a brief overview of the steps taken:

- **Data Collection & Wrangling** Data was gathered using the SpaceX API and web scraping, then cleaned and processed to remove inconsistencies and missing values.
  - Space X API (https://api.spacexdata.com/v4/rockets/)
  - WebScraping (https://en.wikipedia.org/wiki/List_of_Falcon/_9/_and_Falcon_Heavy_launches)

- **Exploratory Data Analysis (EDA)** The data was explored using visualizations and SQL queries to uncover patterns and relationships that influence landing outcomes.
- **Interactive Visual Analytics** Folium maps and a Plotly Dash dashboard were built to allow dynamic and geographic exploration of the launch data.
- **Predictive Analysis** Multiple machine learning classification models were built, tuned, and evaluated to find the best model for predicting successful landings.

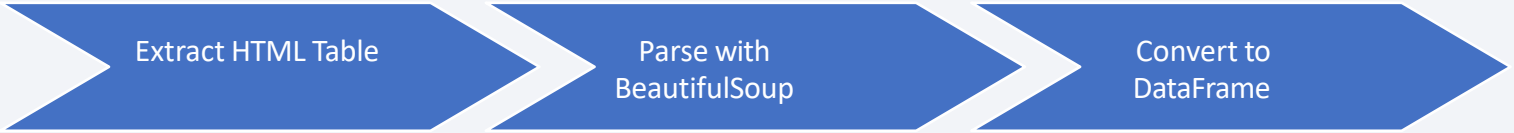
Data Collection

Data was collected from the following:

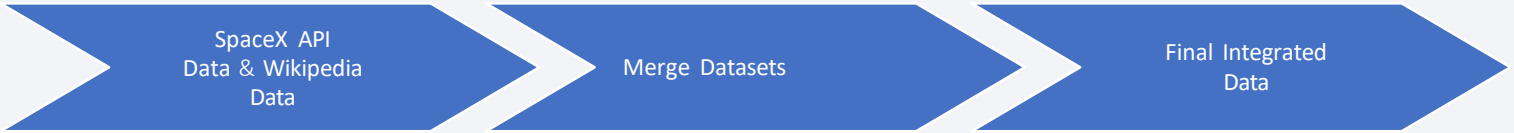
- Space X API (https://api.spacexdata.com/v4/rockets/)
- WebScraping (https://en.wikipedia.org/wiki/List_of_Falcon/_9/_and_Falcon_Heavy_launches)

- Step 1: SpaceX API Request

- Step 2: Web Scraping Wikipedia



- Step 3: Data Integration



Data Collection – SpaceX API

Step 1: Initiate API Request

- Use Python's `requests` library to connect to the SpaceX API.
- Endpoint: `https://api.spacexdata.com/v4/launches`

Step 2: Parse API Response

- Convert API response from JSON to a Python dictionary.
- Extract relevant fields: launch date, launch site, payload mass, rocket type, outcome.

Step 3: Store Data Locally

- Save extracted data into a pandas DataFrame.
- Store the DataFrame locally for further processing.

GitHub URL: 1. jupyter-labs-spacex-data-collection-api.ipynb

Data Collection - Scraping

Step 1: Initiate Web Scraping

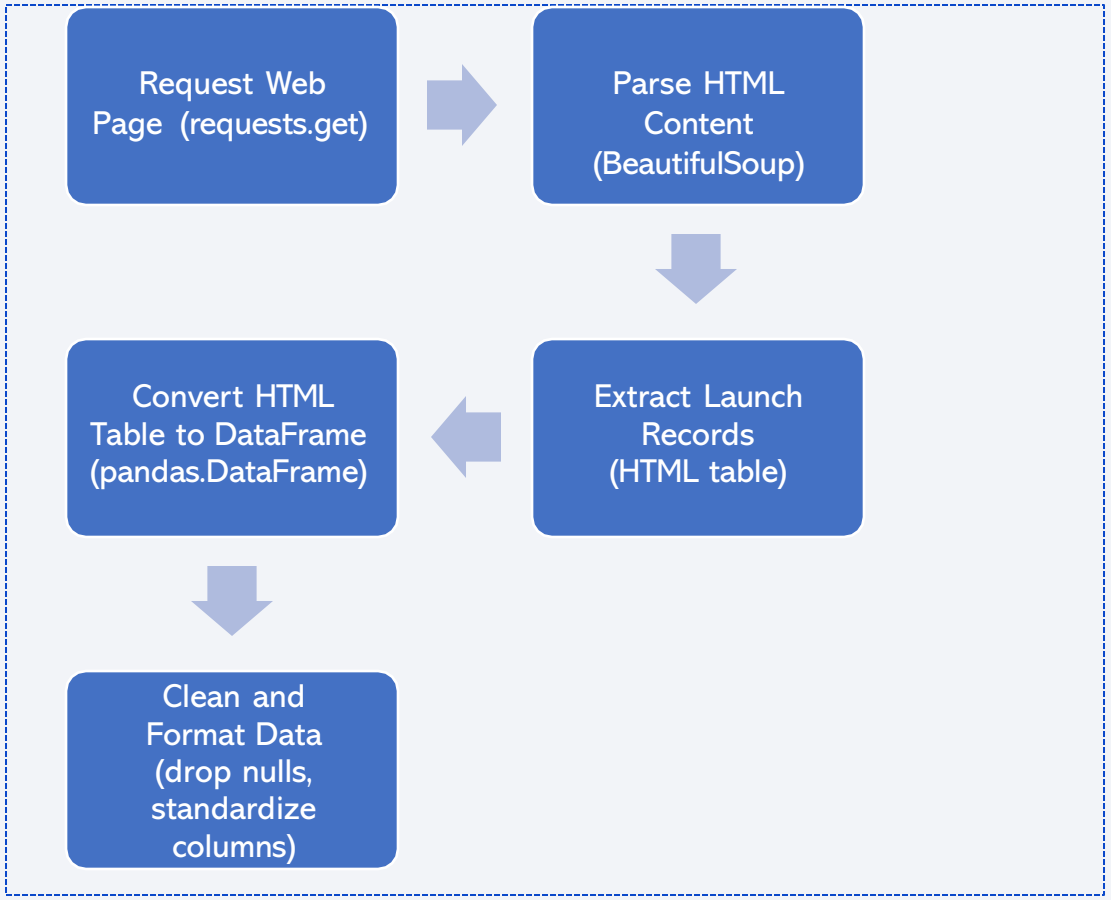
- Use Python's `requests` library to fetch the HTML content of the Wikipedia page.
- Target URL: `https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches`

Step 2: Parse HTML Content

- Use `BeautifulSoup` to parse the HTML content.
- Extract the HTML table containing Falcon 9 launch records.

Step 3: Convert to DataFrame

- Convert the extracted HTML table into a pandas DataFrame.
- Clean and format the DataFrame, ensuring data consistency.



GitHub URL: 2. jupyter-labs-webscraping.ipynb

Data Wrangling

Overview: Data wrangling involves cleaning, transforming, and organizing raw data into a structured format suitable for analysis.

Step 1 — Data Cleaning

- Found and handled missing values using fillna/dropna
- Removed duplicate records

Step 2 — Data Transformation

- Converted columns to the correct data types (dates, numbers)
- Standardized text formatting (lowercase, remove spaces)
- Created new features from existing columns

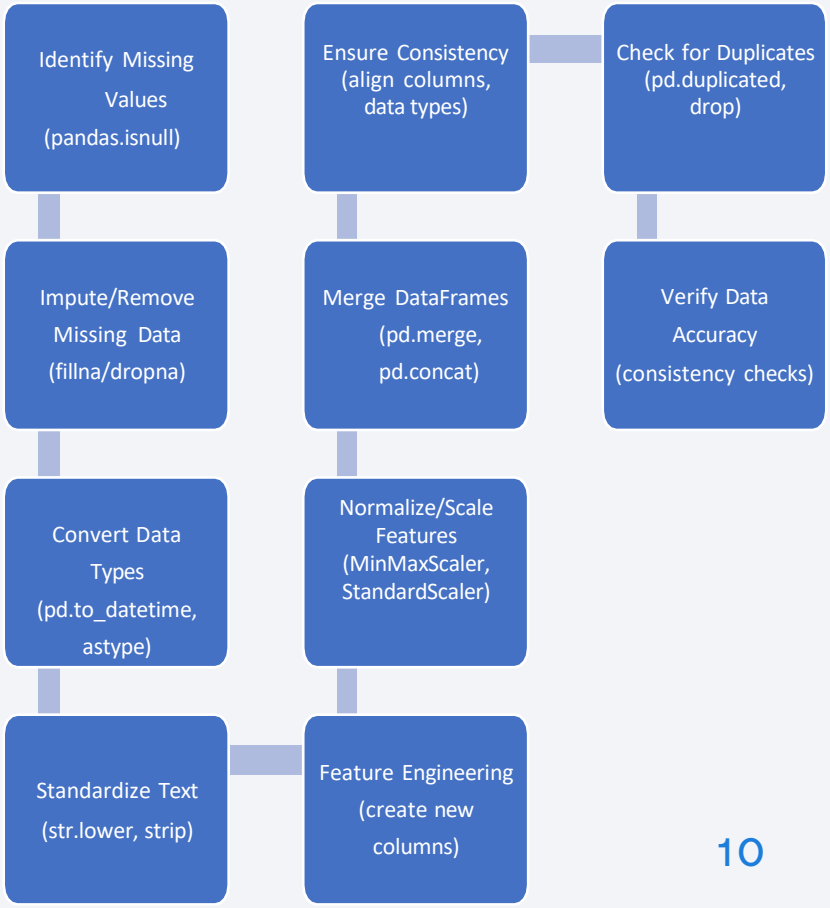
Step 3 — Data Integration

- Merged all data sources into one consistent dataset
- Made sure column names and formats matched across all sources

Step 4 — Data Validation

- Checked for duplicates and removed them
- Verified data accuracy and consistency

GitHub: 3. labs-jupyter-spacex-Data wrangling.ipynb



EDA with Data Visualization

- Overview:

Exploratory Data Analysis (EDA) involves visually exploring and summarizing the main characteristics of a dataset. The goal is to understand the data's distribution, identify patterns, and uncover relationships between variables.

- Histograms — Showed the distribution of payload mass, flight number, and success rates. Helped spot outliers and skewness.
- Bar Charts — Compared launch outcomes across different sites and rocket types. Made patterns easy to see.
- Line Charts — Tracked success rates over time. Revealed improvement trends year by year.Scatter Plots — Explored relationships between payload mass and launch success. Showed correlations between variables.
- Heatmaps — Visualized correlations between multiple numerical variables. Helped with feature selection.
- Box Plots — Displayed data spread and highlighted outliers across categories.

Github URL: 5. jupyter-labs-eda-dataviz.ipynb

EDA with SQL

Aggregate Queries:

- Calculated total number of launches.
- Counted successful and failed launches.
- Calculated success rates by launch site and rocket type.

Join Queries:

- Joined tables to link launch records with additional data (e.g., rocket details).
- Combined datasets for comprehensive analysis.

Filtering Queries:

- Filtered data to focus on specific launch outcomes (success/failure).
- Applied conditions to extract launches based on criteria like launch date or rocket configuration.

Sorting Queries:

- Sorted data to identify trends or outliers.
- Ordered launches by date or success rate for analysis.

Subqueries:

- Nested queries to calculate derived metrics (e.g., average payload mass per launch site).
- Subqueries used to perform detailed analysis within larger datasets.

GitHub URL: 4. jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

Map Objects Created

Markers:

- Placed markers to indicate launch sites on the map.
- Each marker represents a specific geographical location where SpaceX launches have occurred.

Circles:

- Added circles around launch sites to visually represent proximity zones.
- Circles help visualize the areas around launch sites that might influence operational decisions.

Lines:

- Drew lines to connect launch sites with their proximities or other relevant locations.
- Lines provide spatial context and connections between
different points of interest related to launches.

Github URL: 6. lab_jupyter_launch_site_location.ipynb

Reasons for Adding Objects

- To pinpoint exact launch locations for spatial reference.
- Helps users identify where SpaceX has conducted launches geographically.

- Illustrates the potential impact zones around launch sites.
- Provides a visual representation of safety perimeters or operational boundaries.

- Shows connections or relationships between launch sites and relevant features.
- Enhances understanding of spatial relationships and dependencies.

Build a Dashboard with Plotly Dash

Plots/Graphs Added

Success Pie Chart:

- Displays the distribution of successful and failed launches.
- Helps visualize the overall success rate and performance trends.

Success-Payload Scatter Plot:

- Shows the relationship between payload mass and launch success.
- Allows users to explore how payload mass influences mission outcomes.

Github URL: spacex_dash_app.py

Interactions Added

Launch Site Dropdown:

- Enables users to select specific launch sites for analysis.
- Facilitates filtering and focused exploration based on geographical locations.

Range Slider for Payload:

- Allows users to adjust payload mass ranges dynamically.
- Offers flexibility in examining launch success concerning payload mass variations.

Predictive Analysis (Classification)

1. Data Preprocessing:

- Standardized features to ensure all variables contribute equally.
- Split data into training and test sets for model validation.

2. Model Selection:

- Explored multiple classification algorithms: SVM, Decision Trees, and K-Nearest Neighbors (KNN).
- Chose algorithms suitable for binary classification tasks based on project requirements.

3. Hyperparameter Tuning:

- Used GridSearchCV to systematically search for optimal hyperparameters.
- Tuned parameters such as C (SVM), max_depth (Decision Trees), and n_neighbors (KNN).

Github URL:7. SpaceX_Machine_Learning_Prediction_Part_5.jupyterlite.ipynb

4. Model Evaluation:

- Evaluated models using cross-validation techniques to ensure robustness and generalizability.
- Utilized metrics like accuracy, precision, recall, and F1-score to assess model performance.

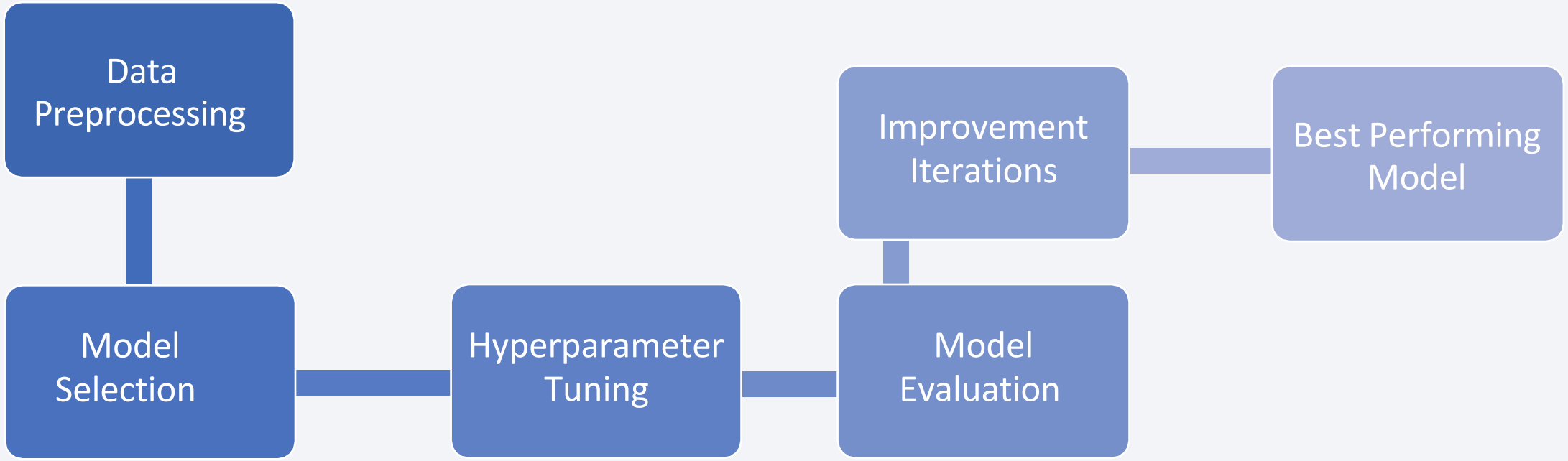
5. Improvement Iterations:

- Iteratively adjusted models based on insights from validation results.
- Fine-tuned hyperparameters to maximize predictive accuracy and reliability.

6. Selection of Best Performing Model:

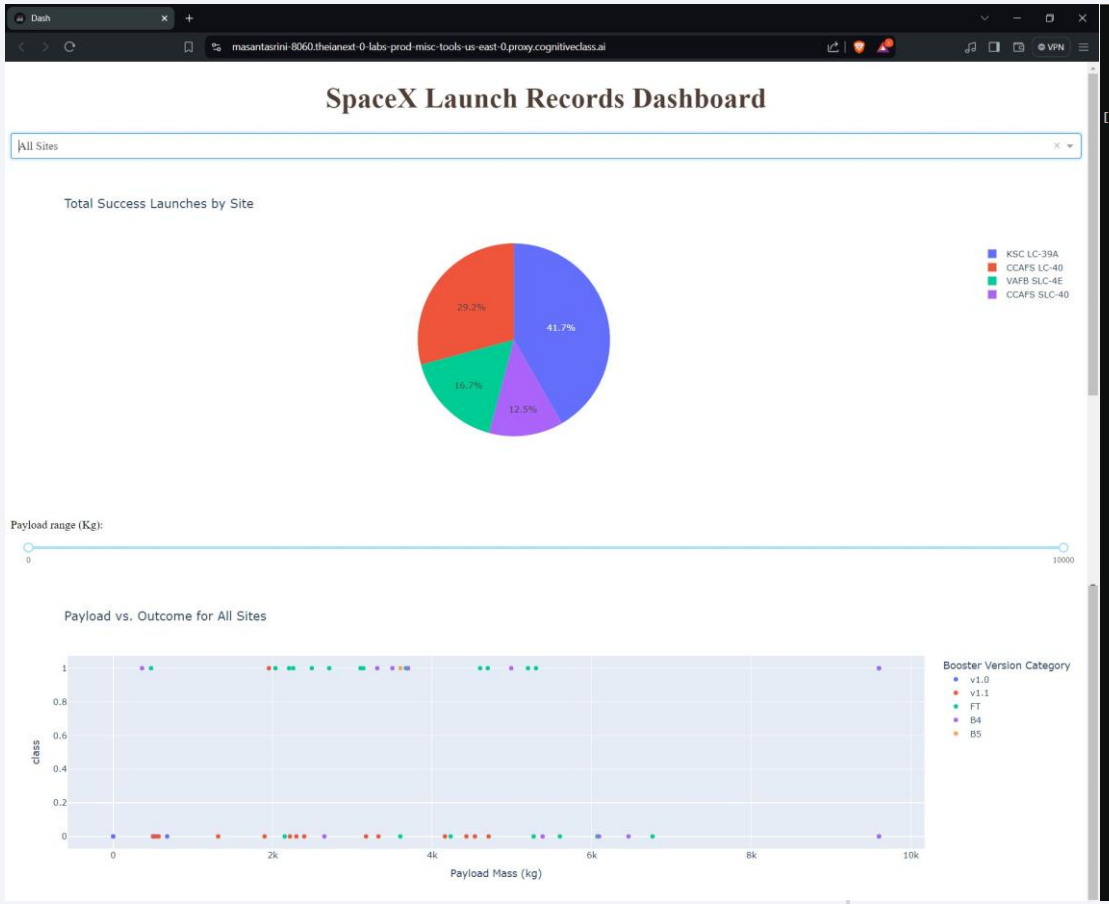
- Identified the model with the highest accuracy on the test set as the best performer.
- Considered both training and test set performance to avoid overfitting and ensure real-world applicability.

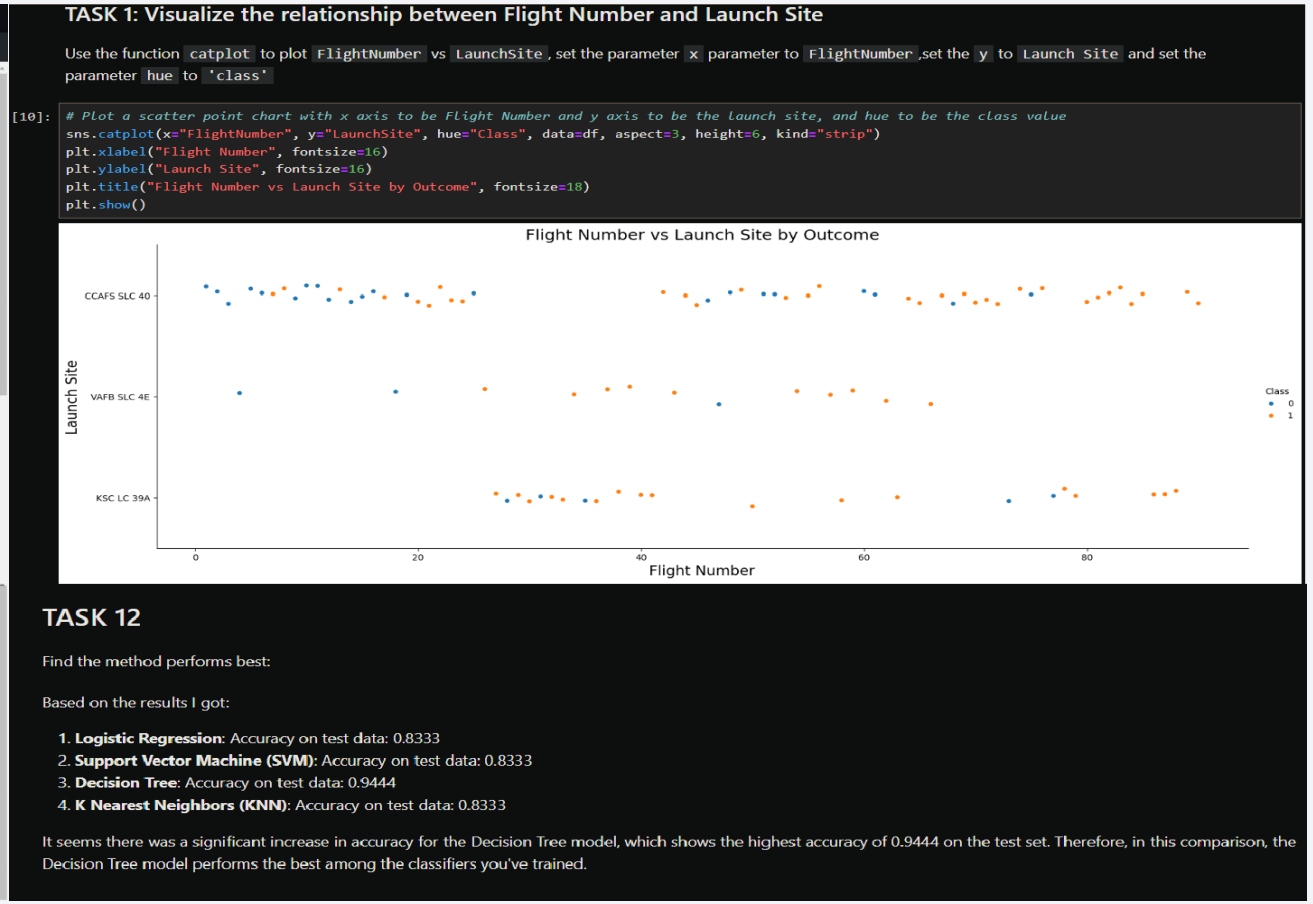
Predictive Analysis (Flowchart)



Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results



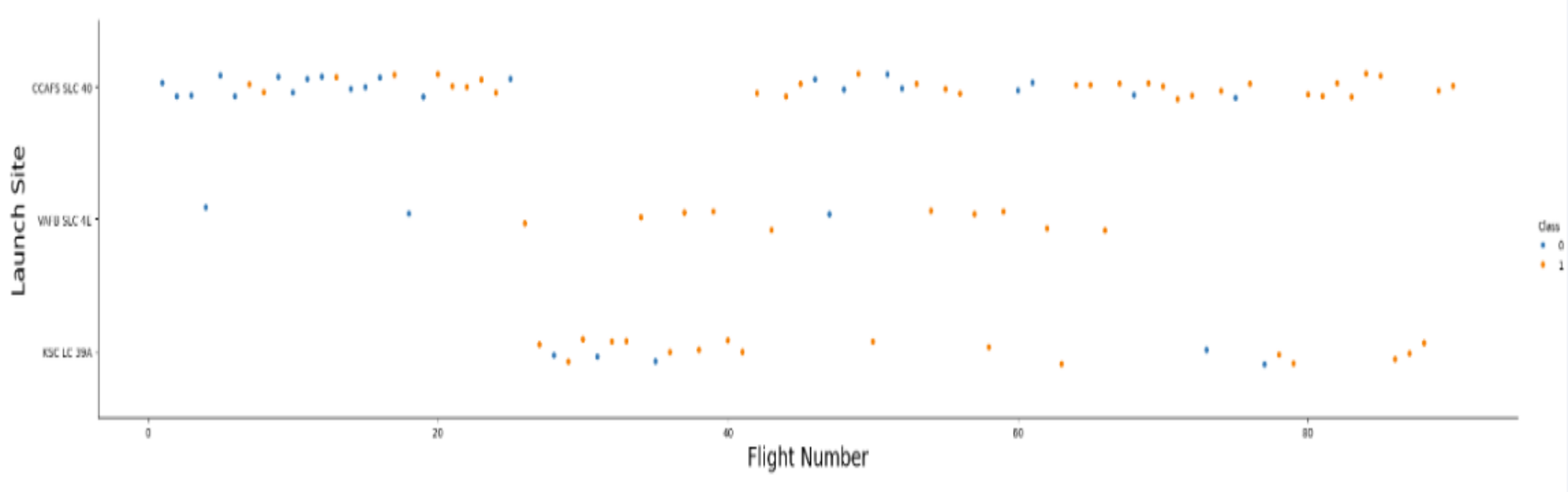


Section 2

Insights drawn from EDA

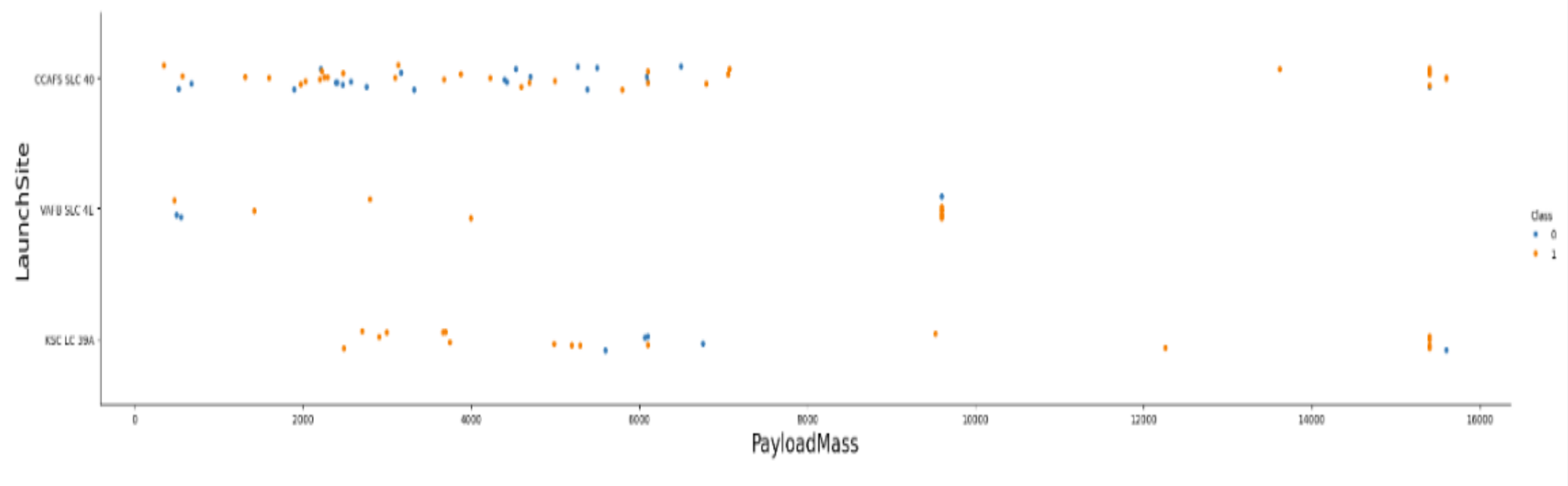
Flight Number vs. Launch Site

- **Mixed Outcomes at Major Launch Sites**: Both CCAFS SLC 40 and KSC LC 39A have a mix of successful (orange) and unsuccessful (blue) landings, indicating that factors other than the launch site itself may influence the landing success.
- **Consistent Activity Across Flight Numbers**: Launches are spread across a wide range of flight numbers at all sites, suggesting consistent activity over time without a clear trend of increasing or decreasing landing success.

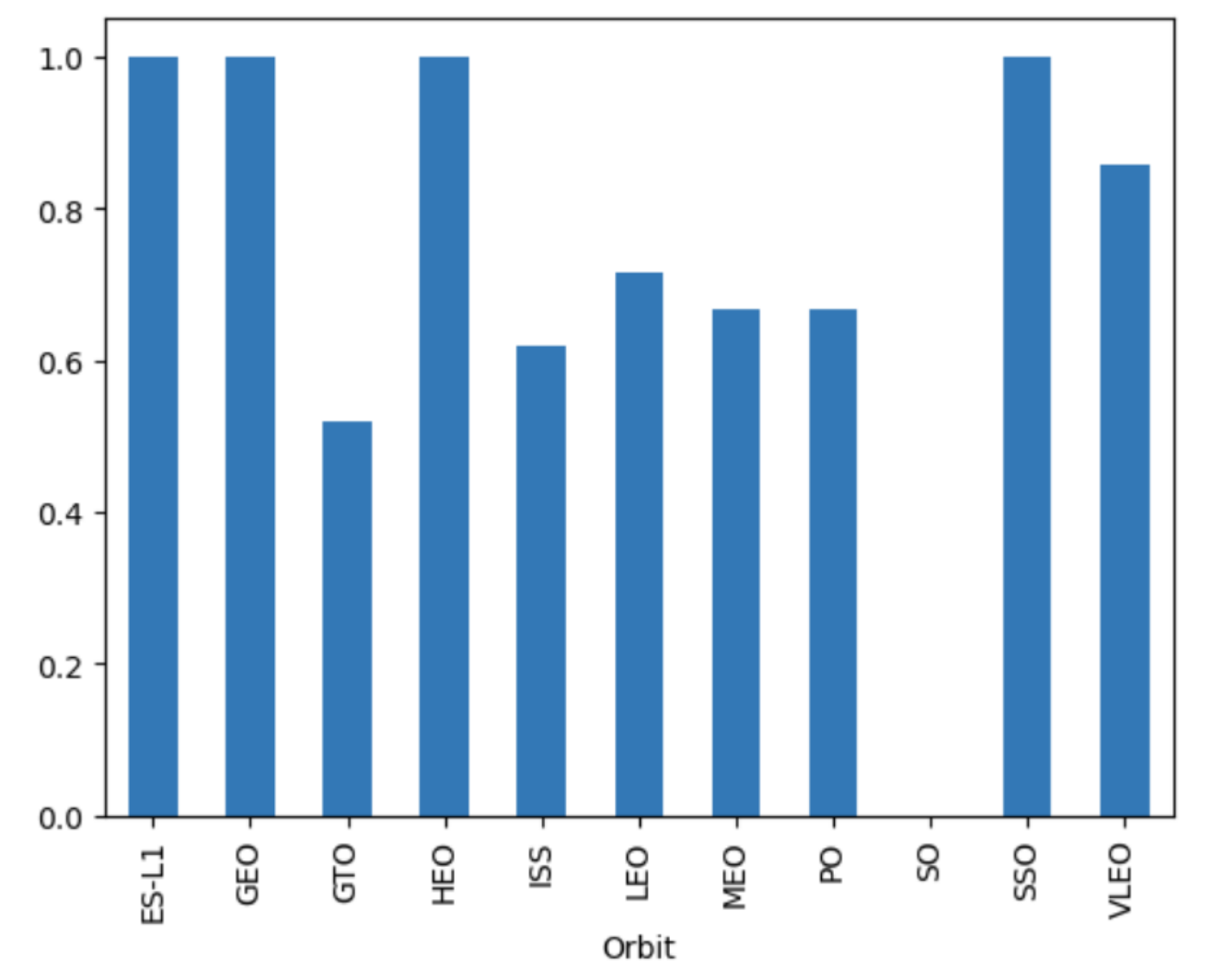


Payload vs. Launch Site

- **Payload Distribution:** Most launches from the CCAFS SLC 40 site handle payloads below 10,000 kg, while the VAFB SLC 4E and KSC LC 39A sites have a wider range of payload masses, indicating varied mission profiles.
- **High-Capacity Launches:** The KSC LC 39A site is frequently used for launching heavier payloads, with multiple launches carrying over 15,000 kg, suggesting its suitability for high-capacity missions.



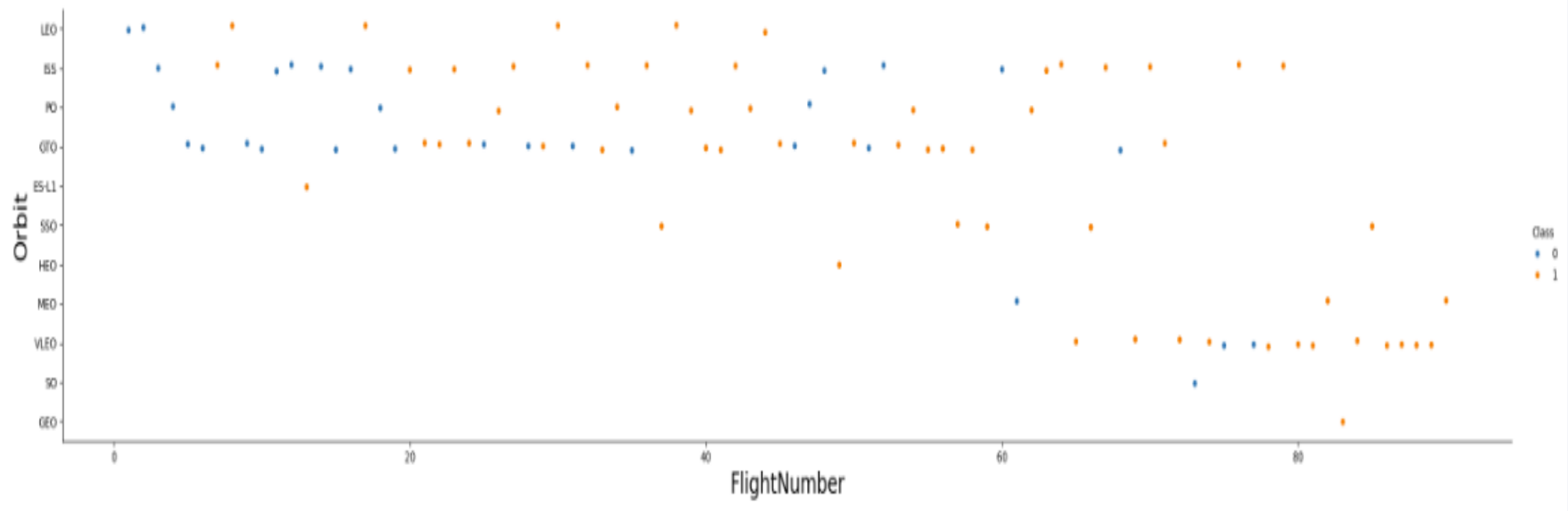
Success Rate vs. Orbit Type



- **High Success Rates:** Missions to VLEO, ES-L1, GEO, HEO, and SSO orbits have achieved a perfect success rate, indicating these orbits are highly reliable for successful first stage landings.
- **Lower Success Rate for GTO:** The GTO orbit type shows a significantly lower success rate compared to other orbit types, suggesting that missions to this orbit may involve greater challenges or complexities.

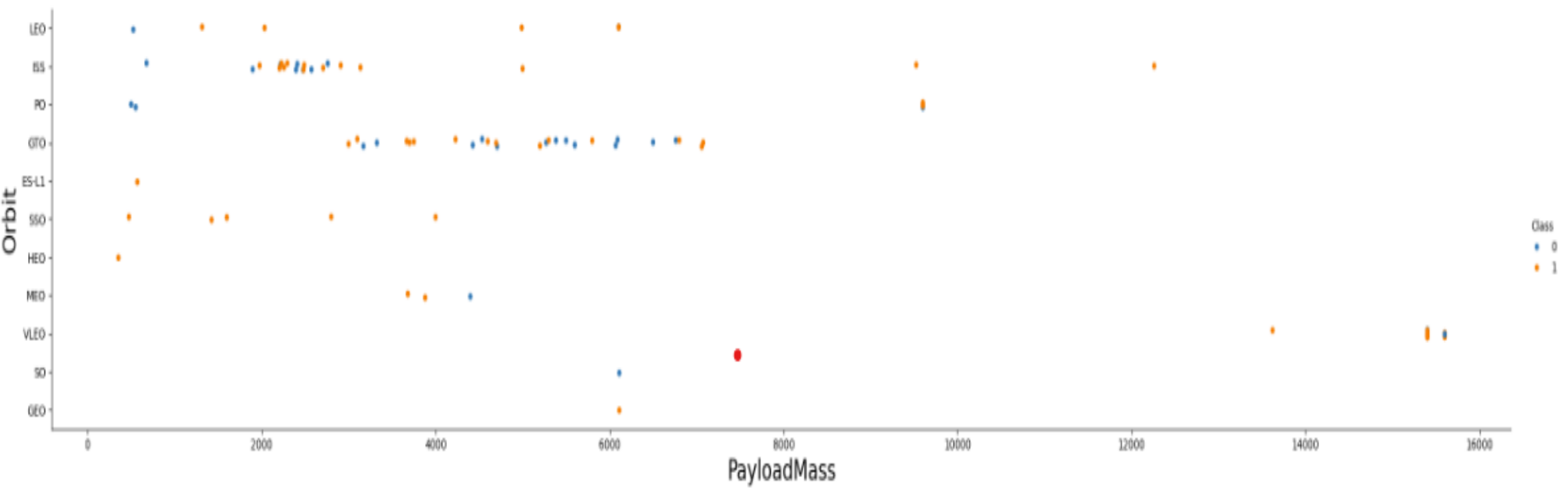
Flight Number vs. Orbit Type

- **Increased Success Over Time:** The success rate of Falcon 9 launches improves significantly with higher flight numbers, indicating that experience and iterative improvements contribute to better outcomes.
- **Orbit-Specific Performance:** Early flights to GTO and ISS orbits had mixed outcomes, but recent missions to these orbits show a higher success rate, reflecting advancements in mission planning and execution.

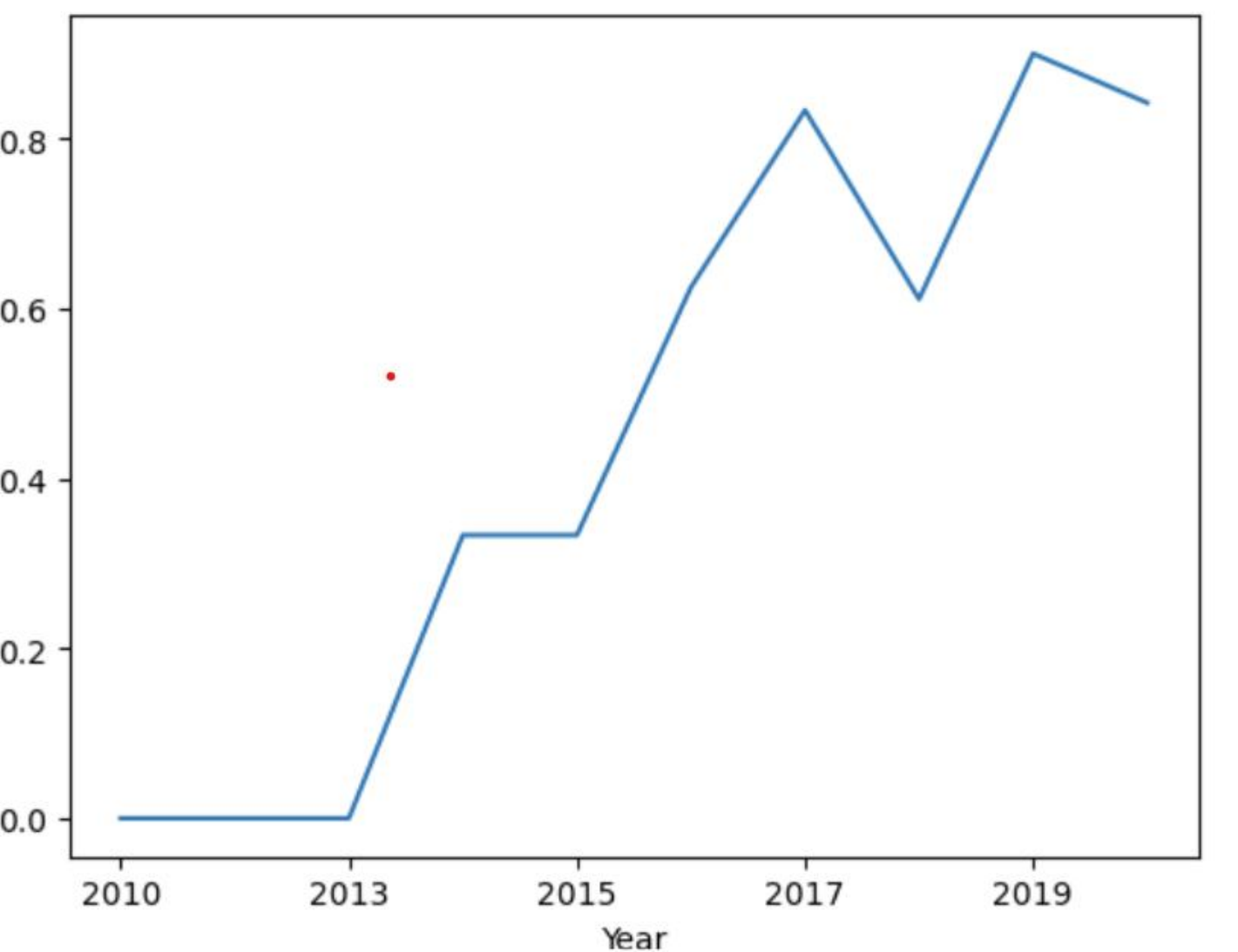


Payload vs. Orbit Type

- With heavy payloads the successful landing or positive landing rate are more for Polar,LEO and ISS.
- However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.



Launch Success Yearly Trend



- You can observe that the sucess rate since 2013 kept increasing till 2020
- Despite a dip in 2018, the overall trend indicates increasing reliability and success in Falcon 9 launches over the years.

All Launch Site Names

Display the names of the unique launch sites in the space mission

[18]:
```
%sql SELECT DISTINCT LAUNCH_SITE as "Launch_Sites" FROM SPACEXTBL;
```

```
 * sqlite:///my_data1.db
Done.
```

[18]:

Launch_Sites
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

In [19]:
```
%sql SELECT * FROM 'SPACEXTBL' WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[19]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

In [20]:
```
%sql SELECT SUM(PAYLOAD_MASS__KG_) as "Total Payload Mass(Kgs)", Customer FROM 'SPACEXTBL' WHERE Customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[20]:

Total Payload Mass(Kgs)	Customer
45596	NASA (CRS)

Average Payload Mass by F9 v1.1

In [21]:
```
%sql SELECT AVG(PAYLOAD_MASS__KG_) as "Payload Mass Kgs", Customer, Booster_Version FROM 'SPACEXTBL' WHERE Booster_Version LIKE 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[21]:

Payload Mass Kgs	Customer	Booster_Version
2928.4	SES	F9 v1.1

First Successful Ground Landing Date

In [22]:
```
%sql SELECT MIN(DATE) FROM 'SPACEXTBL' WHERE "Landing _Outcome" = "Success (ground pad)";
```

```
 * sqlite:///my_data1.db
Done.
```

Out[22]:

MIN(DATE)
None

Successful Drone Ship Landing with Payload between 4000 and 6000

In [22]:
```
%sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS__KG_" > 4000 AND "PAYLOAD_MASS
```

```
 * sqlite:///my_data1.db
Done.
```

Out[22]:

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

In [16]:
```
%sql SELECT "Mission_Outcome", COUNT("Mission_Outcome") as Total FROM SPACEXTBL GROUP BY "Mission_Outcome";
```

```
 * sqlite:///my_data1.db
Done.
```

Out[16]:

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

In [17]:
```
%sql SELECT "Booster_Version",Payload, "PAYLOAD_MASS__KG_" FROM SPACEXTBL WHERE "PAYLOAD_MASS__KG_" = (SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTBL);
```

```
 * sqlite:///my_data1.db
Done.
```

Out[17]:

Booster_Version	Payload	PAYLOAD_MASS__KG_
F9 B5 B1048.4	Starlink 1 v1.0, SpaceX CRS-19	15600
F9 B5 B1049.4	Starlink 2 v1.0, Crew Dragon in-flight abort test	15600
F9 B5 B1051.3	Starlink 3 v1.0, Starlink 4 v1.0	15600
F9 B5 B1056.4	Starlink 4 v1.0, SpaceX CRS-20	15600
F9 B5 B1048.5	Starlink 5 v1.0, Starlink 6 v1.0	15600
F9 B5 B1051.4	Starlink 6 v1.0, Crew Dragon Demo-2	15600
F9 B5 B1049.5	Starlink 7 v1.0, Starlink 8 v1.0	15600
F9 B5 B1060.2	Starlink 11 v1.0, Starlink 12 v1.0	15600
F9 B5 B1058.3	Starlink 12 v1.0, Starlink 13 v1.0	15600
F9 B5 B1051.6	Starlink 13 v1.0, Starlink 14 v1.0	15600
F9 B5 B1060.3	Starlink 14 v1.0, GPS III-04	15600
F9 B5 B1049.7	Starlink 15 v1.0, SpaceX CRS-21	15600

2015 Launch Records

[23]:

Month_Name	Mission_Outcome	Booster_Version	Launch_Site
January	Success	F9 v1.1 B1012	CCAFS LC-40
February	Success	F9 v1.1 B1013	CCAFS LC-40
March	Success	F9 v1.1 B1014	CCAFS LC-40
April	Success	F9 v1.1 B1015	CCAFS LC-40
April	Success	F9 v1.1 B1016	CCAFS LC-40
June	Failure (in flight)	F9 v1.1 B1018	CCAFS LC-40
December	Success	F9 FT B1019	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

[26]:
```
%%sql

SELECT "Landing_Outcome", COUNT(*) AS "Count" FROM   SPACEXTABLE WHERE
    "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY
    "Landing_Outcome" ORDER BY   COUNT(*) DESC;
```

```
 * sqlite:///my_data1.db
Done.
```

[26]:

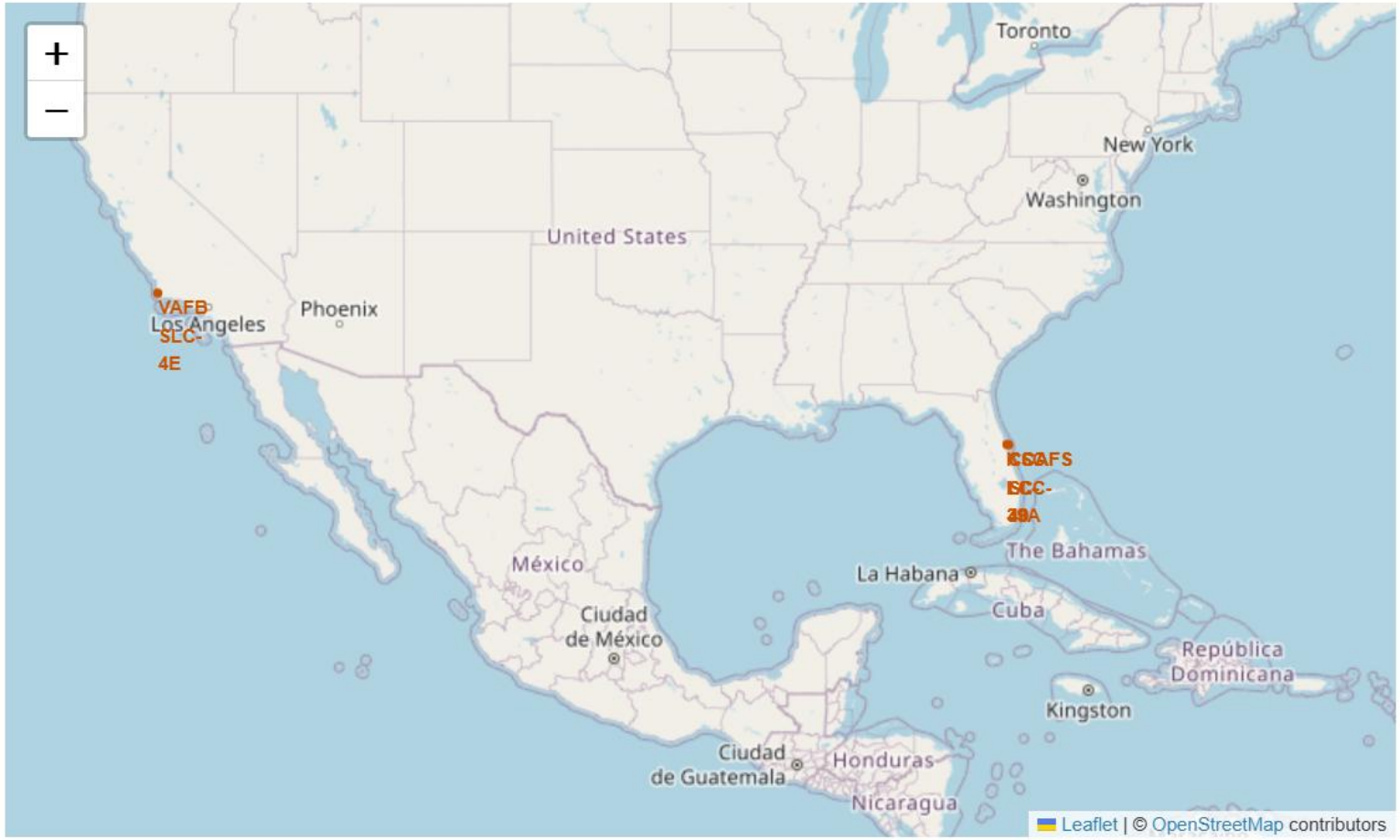
Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis

All launch sites on a map

[11]:



- Launch sites are near sea, probably for safety, but not too far from roads and railroads.

Launch Outcomes by Site

In the provided screenshot, out of 26 launch sites for CCAFS LC-40, there are 19 red markers and 7 green markers. This color-coding helps to quickly identify the success rate and other categorical distinctions of the launches from this specific site. The red markers might represent unsuccessful launches, while the green markers indicate successful ones, providing immediate visual feedback on the performance of launches at each site.

The proximities of launch sites

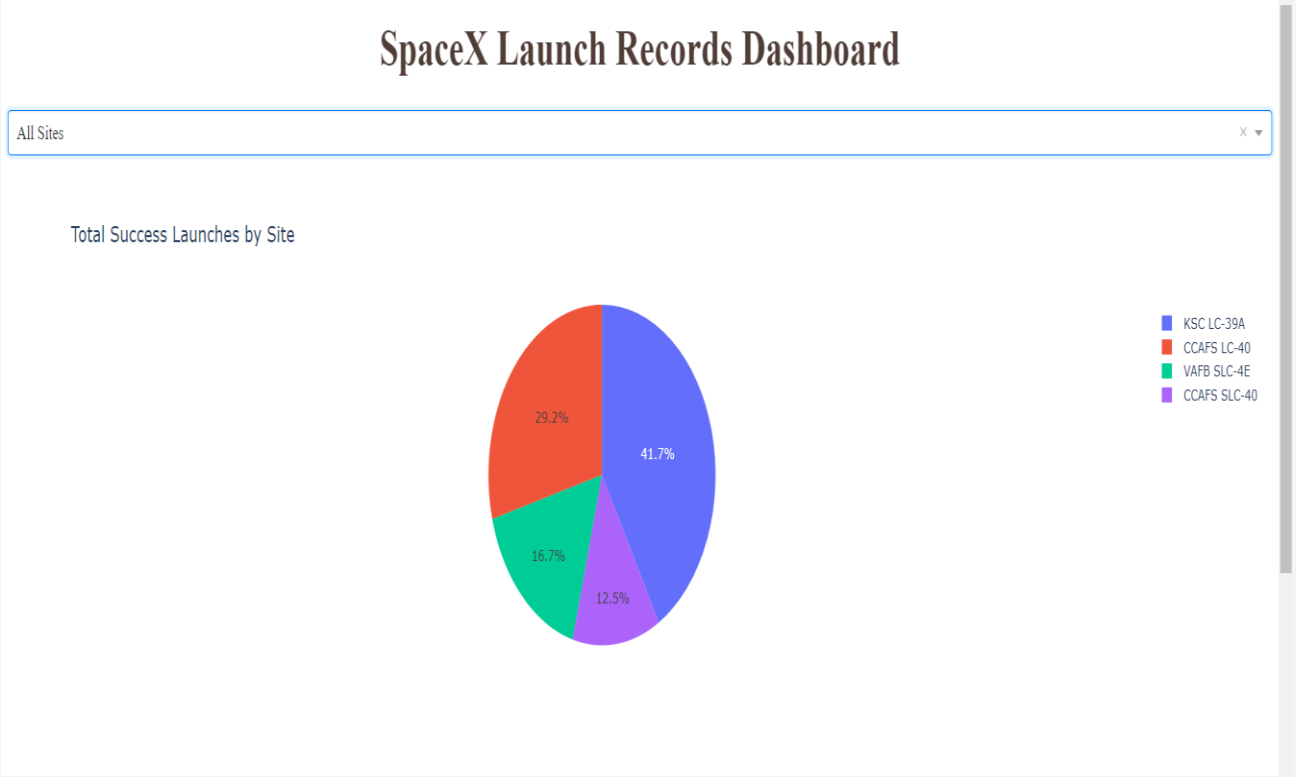
[158]:

This close proximity to the coastline is typical for launch sites to facilitate over-water flight paths and safe recovery operations, ensuring minimal risk to populated areas.

Section 4

Build a Dashboard with Plotly Dash

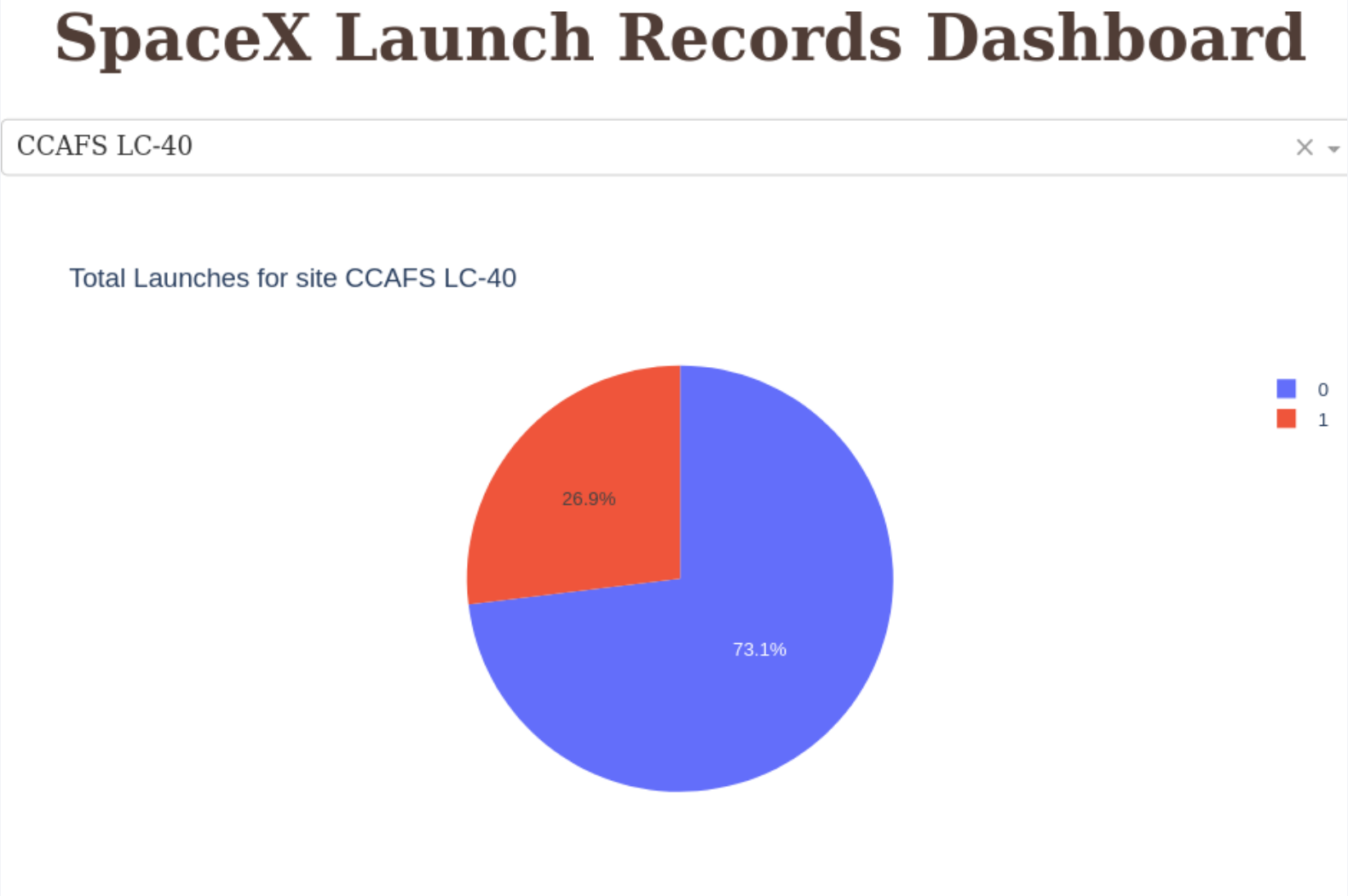
Launch Success Count for all sites (in a pie chart)



Key Findings:

- **CCAFS LC-40**: 29.2%
- **CCAFS SLC-40**: 12.5%
- **VAFB SLC-4E**: 16.7%
- **KSC LC-39A**: 41.7%
- The **KSC LC-39A** launch site has the highest number of successful launches, making up 41.7% of the total successes. This indicates that KSC LC-39A is a highly reliable site for SpaceX launches.

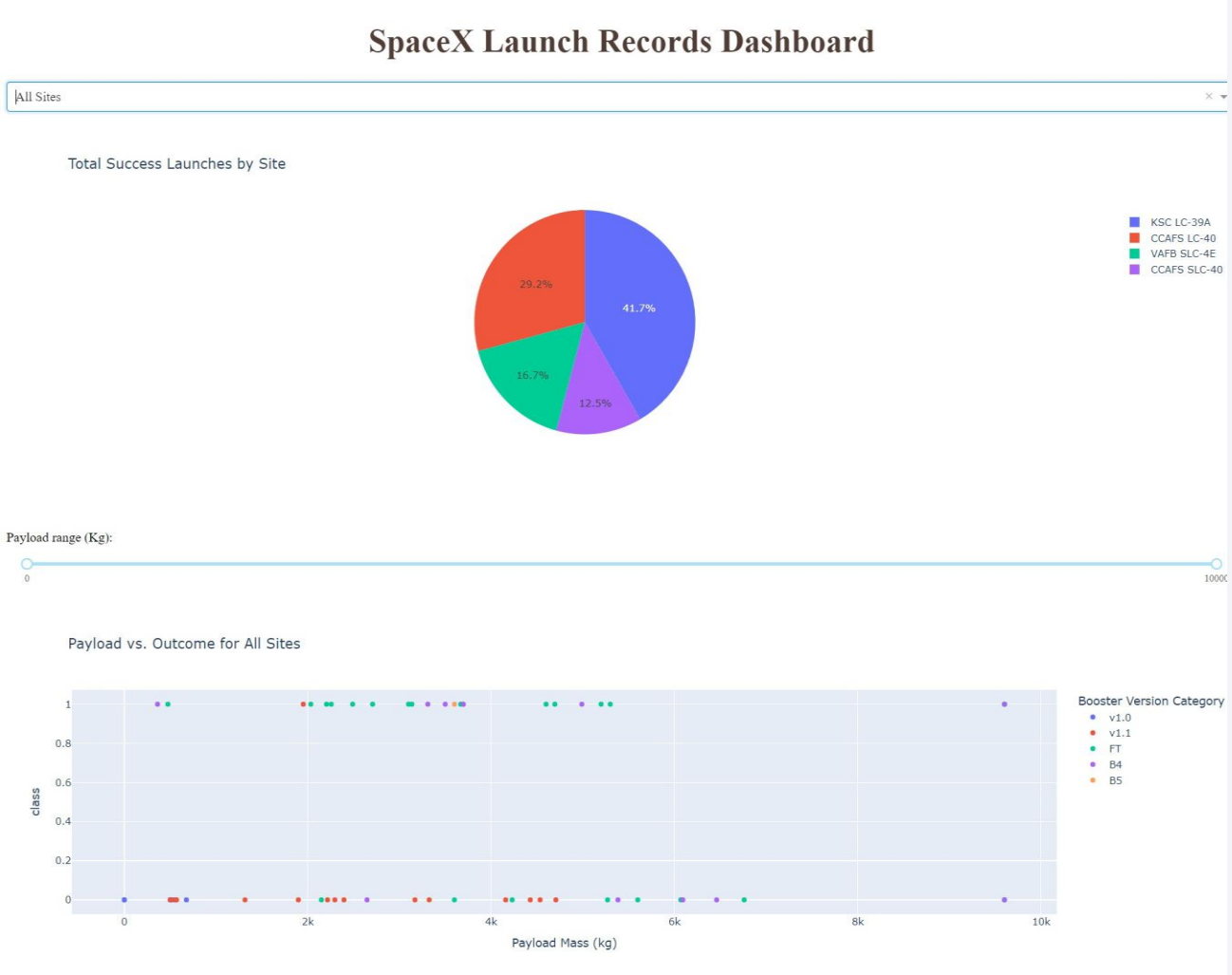
Launch site with highest launch success ratio



Key Findings:

- The significant portion of successful launches from **KSC LC-40** highlights its reliability and effectiveness as a launch site.
- For **KSC LC-40**:
  - **Class 1** (Successful Launches): 73.1%
  - **Class 0** (Unsuccessful Launches): 26.9%

SpaceX Launch Data Dashboard



Launch Site Success Rates:

- **CCAFS LC-40** has the highest success rate with 43.7% of successful launches.
- This suggests that **CCAFS LC-40** is the most reliable launch site among the ones analyzed.
- Other sites like **KSC LC-39A, VAFB SLC- 4E,** and **CCAFS SLC-40** have lower success rates, indicating variability in launch success across different sites.

Booster Version Performance:

- **Booster version "FT"** appears to be the most frequently used and has a high success rate across various payload masses.
- Overall, booster versions do not show a clear trend that higher payload masses correlate with lower success rates.

Section 5

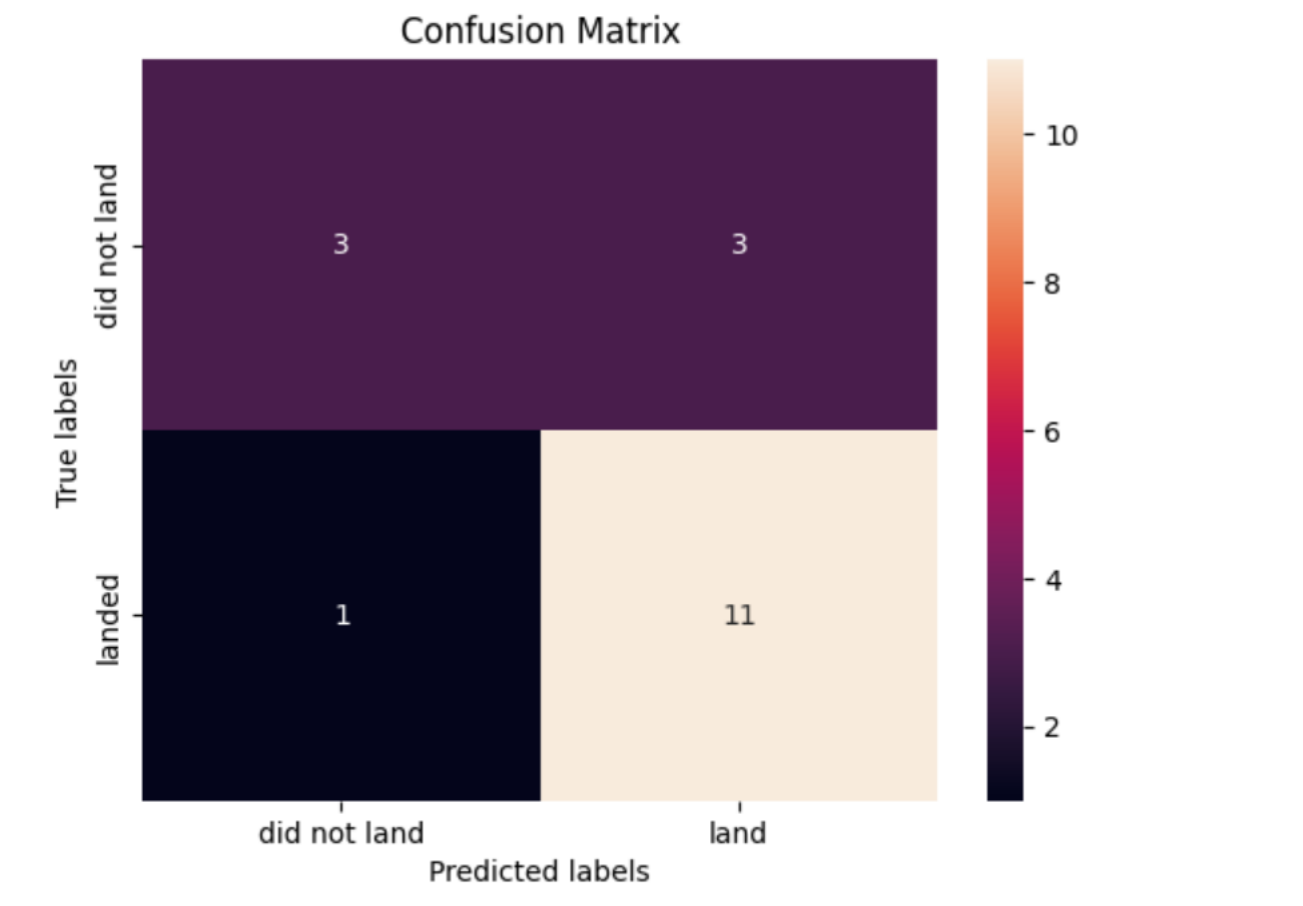
Predictive Analysis (Classification)

Classification Accuracy

```
Logistic Regression Best Accuracy: 0.8464285714285713
Support Vector Machine Best Accuracy: 0.8482142857142856
Decision Tree Best Accuracy: 0.8625
K-Nearest Neighbors Best Accuracy: 0.8482142857142858
The best performing model is Decision Tree with an accuracy of 0.8625
```

- Based on the results, the Decision Tree model has the highest classification accuracy on the test data.
- This suggests that the Decision Tree model is better suited for this dataset compared to Logistic Regression, Support Vector Machine, and K Nearest Neighbors.

Confusion Matrix



Balanced Performance: The model shows a balanced performance with a slight bias towards predicting successful landings. This aligns well with practical needs in the aerospace industry, where ensuring successful landings is of paramount importance for cost estimation and planning.

High Accuracy: The model is strong at catching actual landings — with a 91.7% Recall, it almost never misses a true successful landing (only 1 missed out of 12).

Conclusions

- **CCAFS LC-40** is the top-performing launch site, accounting for ~44% of successful launches, likely due to superior infrastructure and experienced crews.
- The **FT booster version** is the most reliable, consistently achieving high success rates across all payload masses and should be prioritized for future missions.
- **Payload mass alone** does not determine mission success — launch site and booster version have a far greater influence on outcomes.
- **ML models achieved 86% accuracy** in predicting landing outcomes, and should be integrated into pre-launch planning to flag risky configurations early.

Thank you!